MIL-P-43090B(MR)
2 August 1974
SUPERSEDING
MIL-P-43090A(MR)
25 January 1968

# MILITARY SPECIFICATION

## PLASTIC MOLDING MATERIAL, ASPHALT, ASBESTOS FILLED

This specification is approved for use by all departments and Agencies of the Department of Defense.

## 1. SCOPE

1.1 This specification covers one type of asbestos filled asphalt molding compound suitable for molding shipping supports for ammunition items.

## 2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-D-723 - Drums, Fiber
PPP-D-729 - Drums, Shipping and Storage, Steel, 55-Gallon

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

FSC 9330

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing and Materials (ASTM) Standards:

| | |
|---|---|
| D256 | Impact Resistance of Plastics and Electrical Insulating Materials. |
| D570 | Water Absorption of Plastics. |
| D618 | Conditioning Plastics and Electrical Insulating Materials for Testing. |
| D695 | Compressive Properties of Rigid Plastics. |
| D792 | Specific Gravity and Density of Plastics by Displacement. |
| D2240 | Indentation Hardness of Rubber and Plastics by Means of a Durometer. |

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

National Motor Freight Traffic Association, Inc., Agent:

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Associations, Inc., Tariff Order Section, 1616 P Street, N. W., Washington, DC 20036.)

Uniform Classification Committee, Agent:

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, Illinois 60606.)

3. REQUIREMENTS

3.1 Material. The material shall consist of an asbestos filled asphaltic composition.

3.2 Properties. The material shall conform to the property values specified in table I, when tested as specified in the applicable procedure of 4.3.



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Table I. Property values

| Property | Value | Paragraph reference to applicable test |
|---|---|---|
| Specific gravity, 23°/23°C | 1.30 ± 0.05 | 4.3.3 |
| Water absorption, weight gain plus soluble matter loss, percent, max. | 0.19 | 4.3.4 |
| Impact strength, Izod (1/2 inch wide specimens tested at -55°C) foot pounds per inch of notch, min. | 0.55 | 4.3.5 |
| Hardness, Shore D, | | |
| at 23° ± 1°C (73.4°± 1.8°F) | 71 ± 6 | 4.3.6 |
| at 71° ± 2°C (160° ± 3.6°F) | 59 ± 6 | 4.3.6 |
| Compressive strength at rupture, psi, min. | 1800 | 4.3.7 |
| Compressive deformation at 1500 psi, percent | | |
| max. | 4.0 | 4.3.7 |
| min. | 2.5 | 4.3.7 |
| Inorganic residue, percent | 20 ± 7.5 | 4.3.8 |
| Inorganic residue, nature of | Fibrous, resembling asbestos, retaining general shape of original specimen | 4.3.8 |

3.3 Color. Unless otherwise specified by the procuring activity, the color of the material shall be black. If a color other than black is specified, the method of determining compliance, including tolerances, shall be as specified by the procuring activity (see 6.2).

3.4 Suitability for use with explosives. Suitability for use with a particular explosive, when specified by the procuring activity, shall be verified in a Government laboratory. The method of verification shall be in accordance with the preference of the procuring activity (see 6.2 and 6.3).



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3.5 Workmanship. The material shall be clean and free of visible contamination and shall be uniform in color and form.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Sampling for inspection and acceptance. Sampling for inspection shall be performed in accordance with the provisions set forth in MIL-STD-105 except where otherwise indicated. For purposes of sampling, an inspection lot for examination and tests shall consist of all material of the same type, submitted for delivery at one time.

4.2.1 Inspection of materials and components. In accordance with 4.1 above, the supplier is responsible for insuring that materials anc components used were manufactured, examined, and tested in accordance with the requirements of this specification and, to the extent specified, of all referenced subsidiary specifications and standards. In the event of conflict, this specification will govern. A supplier's certificate of compliance with 3.1 shall be furnished.

4.2.2 Inspection of material.

4.2.2.1 Examination of the material. Examination of the material shall be made in accordance with the classification of defects, inspection levels and acceptable quality levels (AQLs) set forth below. The lot size, for purpose of determining the sample size in accordance with MIL-STD-105, shall be expressed in units of 100 pounds for examination in 4.2.2.1.1, and in units of shipping containers for examination in 4.2.2.1.2.

4.2.2.1.1 Examination of the material for defects in appearance, and workmanship. The sample unit for this examination, specified in table II shall be approximately one pound.



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Table II. Examination for defects in appearance and workmanship

| Examine | Defect |
|---|---|
| Appearance and workmanship | Color[1/] not as specified<br>Color not uniform<br>Form not uniform<br>Not clean, presence of foreign material |

[1/] When specified by the procuring activity, (see 3.3 and 3.4).

4.2.2.1.2 Examination of the preparation for delivery. An examination shall be made in accordance with table III, to determine that packaging, packing and marking comply with section 5 requirements. The sample unit for this examination shall be one shipping container fully packed, selected just prior to the closing operation. Shipping containers fully prepared for delivery shall be examined for closure defects.

Table III. Examination of preparation for delivery

| Examine | Defects |
|---|---|
| Packing | Not level specified; not in accordance with contract requirements. |
| | Any nonconforming component, component missing, damaged or otherwise defective affecting serviceability |
| | Inadequate application of components such as: container flaps loose, strappings inadequate, container bulged or distorted. |
| Quantity of material | Less than specified or indicated quantity. |
| Weight | Gross weight exceeds specified requirements. |
| Markings | Interior or exterior markings omitted, illegible, incorrect, incomplete of improper size, location, sequence or method of application. Not in accordance with contract requirements. |



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4.2.2.1.3 Inspection levels and acceptance quality levels (AQLs) for examinations. The inspection levels for determining the sample size and the acceptance quality level (AQL) expressed as defects per 100 units shall be as follows:

| Examination paragraph | Inspection level | AQL |
|---|---|---|
| 4.2.2.1.1 | II | 2.5 |
| 4.2.2.1.2 | S-2 | 2.5 |

4.2.3 Testing. The molding material shall be tested for the applicable characteristics listed in table I in accordance with the test methods specified, for each lot submitted. The lot size, for the purpose of determining the sample size for testing, shall be expressed in units of 100 pounds of material. The sample unit of product to prepare all required specimens shall consist of 10 to 15 pounds. The inspection level shall be S-1, with an acceptance number of 0. The results for each test shall be the averaged results of the specimens.

4.3 Test methods.

4.3.1 Specimen preparation. Test specimens shall be prepared by injection molding in accordance with recommendations of the manufacturer.

4.3.2 Specimen conditioning and testing. Test specimens, excluding that for impact strength, shall be conditioned in accordance with procedure A of ASTM D 618 and tested at 23° ± 2°C (73.4° ± 3.6°F) and 50 ± 5 percent relative humidity.

4.3.3 Specific gravity. Three specimens shall be tested in accordance with ASTM D 792. Method A1 of ASTM D 792 shall be used.

4.3.4 Water absorption. Three specimens shall be tested in accordance with ASTM D 570 using the 24 hour immersion period.

4.3.5 Impact strength. Five specimens shall be tested in accordance with ASTM D 256, using method A. Specimens shall be 1/2 by 1/2 by 2-1/2 inch bar. Specimens shall be conditioned for a minimum of 4 hours at -55° ± 2°C (-67° ± 3.6°F) prior to testing at this temperature.



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4.3.6 Durometer hardness. A minimum of three readings shall be taken equally spaced along the length of each of three 1/2 by 1/2 by 2-1/2 inch bars in accordance with ASTM D 2240. After testing at 23° ± 2°C (73.4° ± 3.6°F), the specimens shall be heated to 71° ± 2°C (160° ± 3.6°F) and tested in the same manner.

4.3.7 Compressive deformation at 1500 psi and compressive strength at rupture. Five 1/2 by 1/2 by 1 inch specimens shall be tested in accordance with ASTM D 695. Both the compressive deformation at 1500 psi stress and the compressive strength at rupture shall be measured on the same specimens.

4.3.8 Inorganic residue. Three specimens of any convenient size shall be weighed in separate crucibles. Initially, the specimens may be subjected to preliminary ashing over a Bunsen Burner in which heat is applied slowly and the contents protected from strong drafts at all times. The specimens shall then be completely ashed in a muffle furnace at 750° ± 25°C (1382° ± 45°F) and the residue of each placed in a desiccator, and allowed to cool. Following cooling, the contents shall be weighed. The ignition, subsequent cooling, and weighing shall be repeated until the weight becomes constant. The inorganic residue shall be expressed as a percentage by weight, and shall be taken as the ratio of the weight of the residue to the weight of the specimen in the condition received, multiplied by 100. The ignited residue shall be examined visually for conformance to the nature of inorganic residue specified in table I.

## 5. PREPARATION FOR DELIVERY

Application. The requirements of section 5 apply only to purchase by or direct shipment to the Government.

5.1 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.1.1 Level A. Unless otherwise specified, the material shall be packed in one of the following types of containers:

(a) Fiber drums conforming to PPP-D-723, type II, grade C, or type III, grade A in quantities of 200 pounds, maximum.

(b) Metal drums conforming to PPP-D-729, type III or type IV, in quantities of 400 pounds, maximum.

Insofar as practical, drums shall be of uniform shape and size, with minimum cube and tare consistent with the protection required. Drums shall contain identical quantities and shall be closed in accordance with the applicable container specification.



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5.1.2 Level B. Unless otherwise specified, the material shall be packed in one of the following types of containers:

(a) Fiber drums conforming to PPP-D-723, type I, grade C in quantities of 200 pounds maximum.

(b) Metal drums conforming to PPP-D-729, type III or type IV, in quantities of 400 pounds, maximum.

Insofar as practical, drums shall be of uniform shape and size with minimum cube and tare consistent with the protection required. Drums shall contain identical quantities and shall be closed in accordance with the applicable container specification.

5.1.3 Level C. The material shall be packed in a manner to ensure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. Containers shall be in accordance with Uniform Freight Classification Rules or National Motor Freight Classification Rules, as applicable.

5.2 Marking. In addition to any special marking specified in the contract or order, containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. This material is intended for use in low cost molded supports for the nose ends of complete rounds of ammunition packed in asphaltic fiber containers during storage and shipment.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Color, other than black, if required (see 3.3).
(c) Suitability for use with explosives, if required (see 3.4).
(d) Level of packing required (see 5.1).

6.3 Suitability for use with explosives. Information on the suitability of many plastics for use with various explosives is on file at Picatinny Arsenal, Dover, New Jersey. Procuring agencies desiring information on the suitability of this material with a particular explosive should first contact Picatinny Arsenal to determine whether the information is already available.

Custodians:
Army - MR

Review interest:
Army - PA, MD
DSA - GS

Preparing activity:
Army - MR

Project No. 9330-A420



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# SPECIFICATION ANALYSIS SHEET

Form Approved Budget Bureau No. 119-R004

**INSTRUCTIONS**

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION: MIL-P-43090B(MR), Plastic Molding Material, Asphalt, Asbestos Filled

| ORGANIZATION | CITY AND STATE |
|---|---|
| | |

| CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT |
|---|---|---|
| | | $ |

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS (*Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.*)

| SUBMITTED BY (*Printed or typed name and activity*) | DATE |
|---|---|
| | |

DD, FORM APR 63 1426 REPLACES NAVSHIPS FORM 4063, WHICH IS OBSOLETE

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